Abdelrahman Khalifa

Embedded Software Engineer | C, 32-bit MCUs, Zephyr RTOS

abdelrahman.khalifa.gad@gmail.com | +20 114 250 5042 | Cairo, Egypt

linkedin.com/in/abd002 | github.com/Abd002 | codeforces.com/profile/_Khalifa

SUMMARY

Embedded software engineer writing C firmware for 32-bit microcontrollers (ESP32-S3, ARM Cortex-M STM32 and TM4C). Owns the Zephyr RTOS firmware for EKSON's pneumatic motion simulators; built a dual-bank STM32 bootloader with CRC/signature check and rollback (graduation project); merged a Zephyr networking contribution. Also works in modern C++, Rust and Python.

EXPERIENCE

Embedded Software Engineer | **EKSON** | **Aug 2025 – Present**

Software for 3-DoF and 6-DoF pneumatic motion simulators: firmware, desktop app, backend and CI

- Wrote the ESP32-S3 firmware and migrated it from ESP-IDF/FreeRTOS to Zephyr RTOS for more deterministic timing, devicetree-based drivers and easier porting to other boards.
- Implemented closed-loop PID control of the pneumatic valves, with feedback from an MPU6050 IMU over I2C (calibration and a Mahony filter for orientation), plus safety stops and a watchdog.
- Developed the PyQt5 desktop app that controls the simulator over a UART protocol; cut its CPU usage and latency by about 40% (measured) using asyncio to wait for ESP32 and server messages concurrently, off the UI thread.
- Engineered remote operation and firmware updates: a Node.js/Express + MongoDB backend that the desktop app polls over REST, then flashes new firmware with esptool; GitHub Actions CI.

Google Summer of Code Contributor | **The Linux Foundation (OpenPrinting)** | **May 2026 – Oct 2026**

Open-source project: Printers page for COSMIC Settings, in Rust | github.com/Abd002/GSOC-2026

- Built the **Printers page for COSMIC Settings** (printer status, supply levels, print queues, adding printers) as 5 Rust crates, with the CUPS/IPP/DNS-SD backend isolated behind a varlink IPC service (cosmic-settings #2182, in review).
- Wrote Rust-to-C FFI bindings (libcups3 DNS-SD) and fixed crashes in libcups' Avahi code; **7 PRs accepted upstream**.

Embedded Linux Systems Intern | **Orange Digital Center (ODC)** | **Dec 2024 – Jan 2025**

- Created custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; worked with device trees, root file systems and hardware flashing. Scored 100% in the final evaluation.

OPEN SOURCE PROJECTS

- **Zephyr RTOS** (Mar 2025): implemented the `IP_MULTICAST_LOOP` socket option for IPv4 across 6 networking modules, reviewed and merged upstream (issue #86977).
- **OpenPrinting libcups and cups-rs** (2026): fixed DNS-SD crashes in libcups (C); ported cups-rs to libcups3 with CI (PRs #18–#22).
- **Rust ecosystem** (Dec 2025): Boa JavaScript engine string optimization (PR #4553) and a Rust Clippy lint fix (PR #16214), both landed upstream.

PROJECTS

- **AUTOSAR-Based V2X Vehicle** (graduation project, STM32): implemented the firmware update path (an ESP32/RPi gateway sends the image over UART) and a dual-bank bootloader with CRC check and rollback. GitHub
- **Seat Heater Control System** (Tiva C, FreeRTOS): real-time controller with temperature monitoring, heater-level control and UART diagnostics, using tasks, mutexes and event flags. GitHub
- **IoT Device Communication System** (C++, Yocto): TCP and UDP multicast between a Pi and QEMU clients. GitHub
- **AUTOSAR PORT and GPIO drivers** (C, TM4C): MCAL drivers developed and tested against the AUTOSAR specification.
- **Kernel Monitor** (C, Linux, ARM, QEMU): character device driver that walks the process list under RCU and exports per-task IDs, memory, threads, context switches and CPU time to user space; run on QEMU. GitHub
- **Door Locker Security System** (C, two MCUs): password lock with keypad, LCD, I2C EEPROM and DC motor; UART between the MCUs. GitHub

TECHNICAL SKILLS

Languages: C, Modern C++, Rust, Python (asyncio, PyQt5), Java, Bash, Verilog

RTOS & firmware: Zephyr (devicetree, west), FreeRTOS, ESP-IDF, OSEK/VDX (Trampoline), 32-bit ARM Cortex-M (STM32, TM4C), ESP32-S3, AVR, AUTOSAR, bootloaders with image verification (CRC/signature), FOTA, PID control

Embedded Linux: Linux, Yocto, Buildroot, kernel modules, device drivers, device trees, QEMU, Raspberry Pi

Protocols & tools: UART, I2C, SPI, CAN, TCP/UDP, REST; Git, GitHub Actions CI, CMake, Make, debugging

Testing: unit and integration (end-to-end) testing, pytest, Python scripting for build and test automation in CI

EDUCATION

Bachelor of Science (B.Sc.), Computer and Systems Engineering | **Minia University** | **2020 – 2025**

- Grade: Very Good with Honors.
- Activities: Codeforces Expert; 4-time ECPC finalist (2021–2024); Treasurer, IEEE Student Branch.